



Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.



## RESULTS

# Results

01 · CB-SEM

confirmatory structural model

**Table 1.** Measurement model (loadings, reliability & item descriptives)

| Construct / Item       | Mean  | SD   | B    | SE   | z     | p     | Std. loading | $\omega$ | $\alpha$ | CR  | AVE |
|------------------------|-------|------|------|------|-------|-------|--------------|----------|----------|-----|-----|
| <i>esg</i>             |       |      |      |      |       |       |              | .86      | .86      | .86 | .60 |
| esg1                   | −0.07 | 1.02 | 1.00 | 0.00 | 0     | <.001 | .83          |          |          |     |     |
| esg2                   | −0.06 | 1.00 | 0.88 | 0.06 | 14.85 | <.001 | .75          |          |          |     |     |
| esg3                   | −0.02 | 0.98 | 0.92 | 0.06 | 15.60 | <.001 | .79          |          |          |     |     |
| esg4                   | −0.01 | 1.00 | 0.86 | 0.05 | 16.74 | <.001 | .73          |          |          |     |     |
| <i>norm</i>            |       |      |      |      |       |       |              | .88      | .88      | .88 | .65 |
| norm1                  | −0.02 | 1.04 | 1.00 | 0.00 | 0     | <.001 | .84          |          |          |     |     |
| norm2                  | 0.02  | 1.02 | 0.91 | 0.05 | 18.02 | <.001 | .78          |          |          |     |     |
| norm3                  | −0.02 | 1.01 | 0.93 | 0.05 | 17.53 | <.001 | .80          |          |          |     |     |
| norm4                  | −0.01 | 1.02 | 0.95 | 0.05 | 17.54 | <.001 | .81          |          |          |     |     |
| <i>service_quality</i> |       |      |      |      |       |       |              | .86      | .86      | .86 | .62 |
| service_quality1       | −0.03 | 0.99 | 1.00 | 0.00 | 0     | <.001 | .80          |          |          |     |     |
| service_quality2       | −0.01 | 0.96 | 0.89 | 0.07 | 13.42 | <.001 | .74          |          |          |     |     |
| service_quality3       | −0.02 | 0.97 | 1.01 | 0.06 | 16.49 | <.001 | .83          |          |          |     |     |
| service_quality4       | 0.00  | 1.00 | 0.96 | 0.06 | 15.51 | <.001 | .77          |          |          |     |     |
| <i>attitude</i>        |       |      |      |      |       |       |              | .88      | .88      | .88 | .65 |
| attitude1              | 0.05  | 0.98 | 1.00 | 0.00 | 0     | <.001 | .82          |          |          |     |     |
| attitude2              | 0.04  | 1.04 | 1.08 | 0.06 | 19.49 | <.001 | .84          |          |          |     |     |
| attitude3              | 0.08  | 1.01 | 0.98 | 0.06 | 16.47 | <.001 | .78          |          |          |     |     |
| attitude4              | 0.07  | 0.98 | 0.97 | 0.05 | 18.52 | <.001 | .80          |          |          |     |     |
| <i>intent</i>          |       |      |      |      |       |       |              | .84      | .84      | .84 | .65 |
| intent1                | −0.05 | 0.96 | 1.00 | 0.00 | 0     | <.001 | .83          |          |          |     |     |
| intent2                | −0.02 | 0.98 | 0.94 | 0.06 | 14.64 | <.001 | .76          |          |          |     |     |
| intent3                | −0.08 | 0.99 | 1.01 | 0.06 | 15.71 | <.001 | .81          |          |          |     |     |
| <i>null</i>            |       |      |      |      |       |       |              | —        | —        | —   | —   |
| norm1.attitude1        | —     | —    | 1.00 | 0.00 | 0     | <.001 | .68          |          |          |     |     |
| norm2.attitude2        | —     | —    | 0.98 | 0.13 | 7.84  | <.001 | .63          |          |          |     |     |
| norm3.attitude3        | —     | —    | 0.97 | 0.13 | 7.64  | <.001 | .65          |          |          |     |     |
| norm4.attitude4        | —     | —    | 0.99 | 0.16 | 6.22  | <.001 | .64          |          |          |     |     |

**Item sample:** Item Mean/SD are computed on the listwise estimation sample (the same N as the model fit).

**Table 2.** Discriminant validity (Fornell–Larcker)

|                 | esg    | norm   | service_quality | attitude | intent |
|-----------------|--------|--------|-----------------|----------|--------|
| esg             | .78    |        |                 |          |        |
| norm            | -.11   | .81    |                 |          |        |
| service_quality | .00    | -.14*  | .79             |          |        |
| attitude        | -.02   | -.03   | .16**           | .81      |        |
| intent          | .37*** | .26*** | .23***          | .32***   | .80    |

\*p<.05, \*\*p<.01, \*\*\*p<.001

**Table 3.** Discriminant validity (HTMT)

|                 | esg | norm | service_quality | attitude | intent |
|-----------------|-----|------|-----------------|----------|--------|
| esg             |     |      |                 |          |        |
| norm            | .10 |      |                 |          |        |
| service_quality | .05 | .13  |                 |          |        |
| attitude        | .03 | .03  | .14             |          |        |
| intent          | .37 | .26  | .23             | .31      |        |

**Discriminant validity:** Discriminant validity also has its own card (AVE / convergent validity); it is included here so one run gives the complete measurement-model writeup.

**Table 4.** Fit indices

| $\chi^2$ (df, p)   | $\chi^2$ /df | CFI  | TLI  | RMSEA [90% CI] | SRMR |
|--------------------|--------------|------|------|----------------|------|
| 226.74 (215, .278) | 1.05         | 1.00 | 1.00 | .01 [.00, .02] | .03  |

**Table 5.** Structural paths, indirect effects & moderation

| H            | Path                                        | B    | Std. $\beta$ | p     | Percentile 95% CI |       | BC 95% CI |       | Result    |
|--------------|---------------------------------------------|------|--------------|-------|-------------------|-------|-----------|-------|-----------|
|              |                                             |      |              |       | Lower             | Upper | Lower     | Upper |           |
| Direct paths |                                             |      |              |       |                   |       |           |       |           |
| H1           | esg $\rightarrow$ intent                    | 0.41 | .43          | <.001 | .32               | .50   | .32       | .50   | Supported |
| H2           | norm $\rightarrow$ intent                   | 0.30 | .32          | <.001 | .20               | .39   | .20       | .39   | Supported |
| H3           | service_quality $\rightarrow$ intent        | 0.23 | .23          | <.001 | .13               | .33   | .13       | .33   | Supported |
| Moderation   |                                             |      |              |       |                   |       |           |       |           |
| H4           | norm $\rightarrow$ intent $\times$ attitude | 0.30 | .26          | <.001 | .19               | .46   | .18       | .44   | Supported |

**Indirect effects:** The indirect-effects section of Table 5 appears only when the drawn structural paths form a chain (X → M → Y); each indirect effect is a lavaan defined effect with a bootstrapped 95% CI.

**CIs:** Bias-corrected CIs benefit from ≥7,000 resamples (Andrews & Buchinsky, 2000); consider the 10,000 publication-grade preset for final runs.

**Table 6.** Conditional effects (simple slopes)

| Moderator level | B    | SE   | p     | boot 95% CI |
|-----------------|------|------|-------|-------------|
| -1SD            | 0.05 | 0.07 | .474  | [-.10, .19] |
| mean            | 0.30 | 0.05 | <.001 | [.20, .39]  |
| +1SD            | 0.54 | 0.07 | <.001 | [.40, .69]  |

**Scope:** Tables shown follow the pipeline stages you ran (EFA → CFA → fit → structural); if EFA was deselected, the E1/E2 preamble is omitted; if the structural stage was deselected, Table 5 is omitted.

**Cutoffs:** Good-fit guidelines (Hu & Bentler, 1999; Marsh, Hau & Wen, 2004): CFI/TLI ≥ .95, RMSEA ≤ .06 [90% CI], SRMR ≤ .08 - guidelines, not pass/fail gates; RMSEA is unstable at small df / small N, so interpret it cautiously for compact models.

**Estimator:** Use WLSMV for ordinal indicators.

**R²:** R² is filled once per endogenous (outcome) construct. R²(intent) = .46

**Caution:** EFA on the same sample is exploratory - treat it as a diagnostic, not confirmatory evidence.

**Moderation:** Moderation edges appear only when drawn on the canvas, each an interaction-term effect from the same bootstrap run; simple slopes at -1 SD / mean / +1 SD are estimated as defined parameters.

Figure. Model

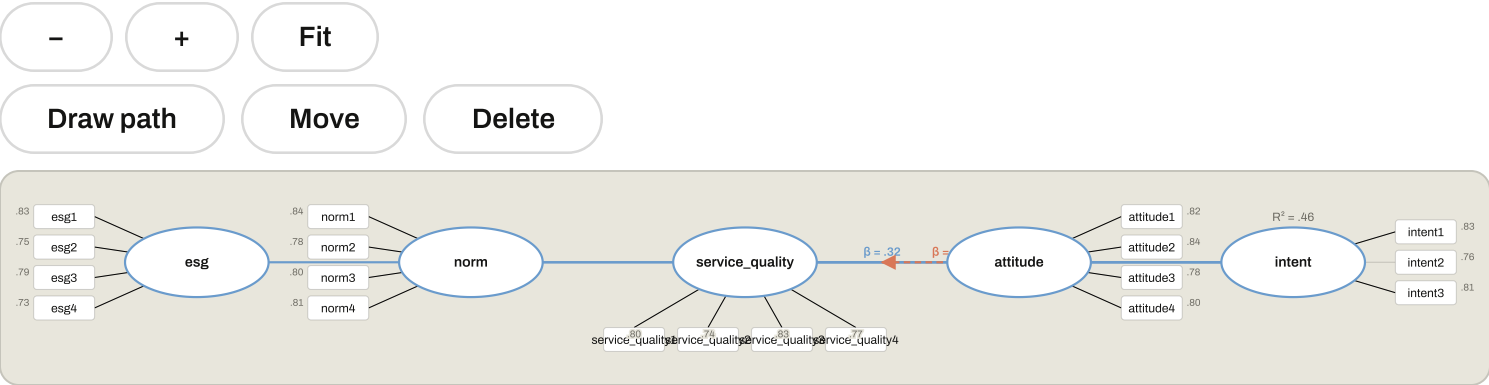
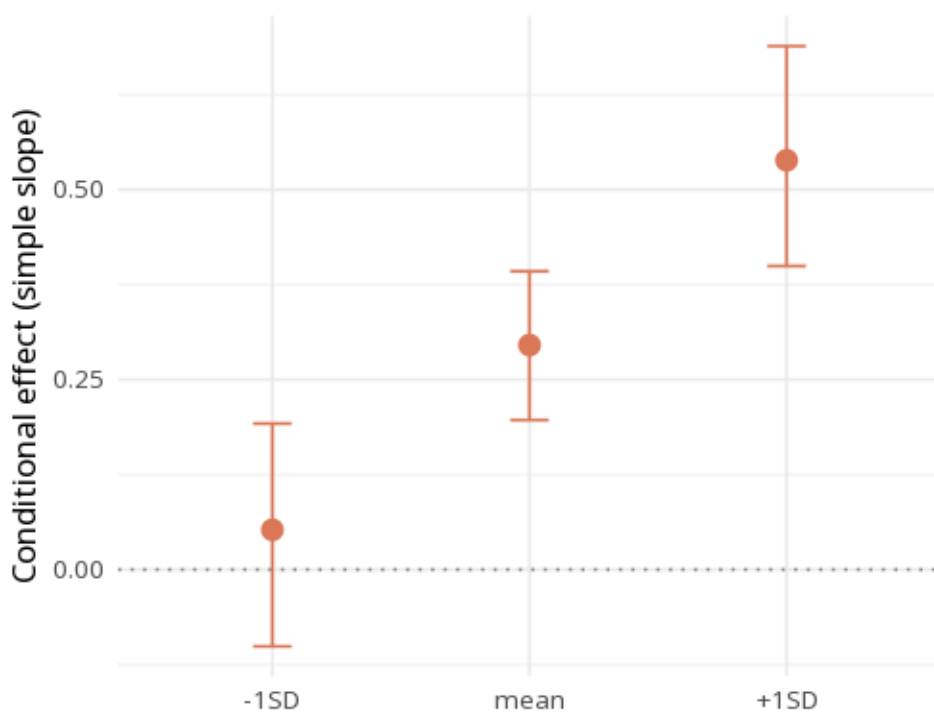


Figure. Simple slopes



## Understanding the numbers

**CFI.** How much better the model fits than a baseline with no relationships at all ( $\geq .95$  is a common, non-binding guideline). Here, CFI = 1.00.

**RMSEA.** The average model misfit per degree of freedom, penalizing complexity ( $\leq .06$  is a common, non-binding guideline). Here, RMSEA = .01 [90% CI .00, .02].

**TLI.** Like CFI, compares the model to a baseline with no relationships, penalizing complexity more than CFI does ( $\geq .95$  is a common, non-binding guideline). Here, TLI = 1.00.

**SRMR.** The average standardized difference between the observed and model-implied correlations ( $\leq .08$  is a common, non-binding guideline). Here, SRMR = .03.

**$\chi^2$  (model fit).** Tests whether the model-implied covariances differ from the observed ones; RMSEA is unstable at small df/small N, so this and the other fit indices are read together, not in isolation. Here,  $\chi^2(215) = 226.74$ ,  $p = .278$ .

**$\chi^2/\text{df}$ .** The chi-square divided by its degrees of freedom - a size-adjusted view of the same model-fit discrepancy. Here,  $\chi^2/\text{df} = 1.05$ .

**Mean.** An item's own observed average, shown for descriptive context alongside its loading. Here, item means range from  $-0.08$  to  $0.08$  across the 23 items in the measurement-model table.

**SD.** An item's own observed standard deviation. Here, item SDs range from  $0.96$  to  $1.04$  across the 23 items above.

**B.** The unstandardized estimate for whichever row you're reading - a factor loading, a structural path coefficient, or (under moderation) a simple slope. Here, factor-loading B's range from  $0.86$  to  $1.08$  across the 23 items; the structural-paths and conditional-effects tables each report their own B per row.

**SE.** The standard error of the estimate in whichever row you're reading - a factor loading or a simple slope. Here, factor-loading SEs range from 0.00 to 0.16 across the 23 items; the conditional-effects table reports its own SE per moderator level.

**z.** A loading's estimate divided by its SE - how many standard errors it is from zero. Here, loading z-values range from 0 to 19.49 across the 23 items.

**p.** The significance of the estimate in whichever row you're reading - a loading, a structural path, or a moderation edge. Here, factor-loading p-values range from <.001 to <.001 across items, and structural-path p-values range from <.001 to <.001; the merged table's own Result column is instead derived from whether the bootstrap CI excludes zero, not from p.

**Std. loading.** The loading rescaled so item variances are comparable across the model. Here, standardized loadings range from .63 to .84 across the 23 items.

**$\omega$  (omega).** Model-based reliability for a construct, reported once per construct alongside its loadings. Here,  $\omega$  ranges from .84 to .88 across the 5 constructs.

**$\alpha$  (alpha).** Cronbach's alpha for a construct, reported alongside  $\omega$ . Here,  $\alpha$  ranges from .84 to .88 across the 5 constructs.

**CR.** Composite reliability for a construct - part of confirming the measurement model before reading the structural paths (this card's own how-to-read). Here, CR ranges from .84 to .88 across the 5 constructs.

**AVE.** Average variance extracted for a construct - part of the same CR/AVE adequacy check. Here, AVE ranges from .60 to .65 across the 5 constructs.

**H.** Which hypothesis number a row of the merged table reports - direct paths first, then indirect effects, then moderation, each numbered in the order you drew or defined them. Here, this run reports 4 H-numbered rows.

**Std.  $\beta$ .** The standardized structural-path coefficient, comparable across paths regardless of each variable's own units. Here, standardized path coefficients range from .23 to .43 across the 4 rows.

**Percentile 95% CI - lower bound.** The lower bound of the primary bootstrap confidence interval for a row. Here, percentile-CI lower bounds range from .13 to .32 across the structural paths.

**Percentile 95% CI - upper bound.** The upper bound of the primary bootstrap confidence interval for a row. Here, percentile-CI upper bounds range from .33 to .50 across the structural paths.

**BC 95% CI - lower bound.** The lower bound of the bias-corrected bootstrap interval, shown alongside the percentile CI. Here, bias-corrected lower bounds range from .13 to .32 across the structural paths.

**BC 95% CI - upper bound.** The upper bound of the bias-corrected bootstrap interval. Here, bias-corrected upper bounds range from .33 to .50 across the structural paths.

**Result.** Whether a row's percentile 95% bootstrap CI excludes zero (Supported) or straddles it (Not supported), at  $\alpha = .05$ . Here, 4 of 4 rows read Supported.

**Moderator level.** Which point on the moderator's distribution a simple slope is estimated at: -1 SD, the mean, or +1 SD. Here, this run estimates simple slopes across 1 moderation edge(s), each at the three fixed levels (-1 SD, mean, +1 SD).

**Boot 95% CI.** The bootstrap 95% confidence interval for a simple slope, in the conditional-effects table. Here, conditional-effect CI bounds range from -.10 to .69 across the estimated moderator levels.

**Factor 1 loading.** An item's loading on the first rotated factor from the optional EFA-preamble stage. Here, Factor 1 loadings appear in the E2 preamble table when the EFA stage is run.

**Factor 2 loading.** An item's loading on the second rotated factor from the optional EFA-preamble stage. Here, Factor 2 loadings appear alongside Factor 1 in the E2 preamble table when the EFA stage is run.

**Communality.** How much of an item's variance is explained by the retained EFA-preamble factors together. Here, communalities appear in the E2 preamble table when the EFA stage is run.

### How to read this test

First confirm the measurement model (loadings high, CR/AVE adequate) and overall fit indices. Then read the structural paths: each std.  $\beta$  with p/CI is a hypothesized relationship between constructs;  $R^2$  shows variance explained in each outcome construct. CB-SEM is confirmatory: the measurement model must be specified from theory a priori - any post-hoc respecification (e.g. from modification indices) is exploratory, must be reported as such, and ideally cross-validated on a fresh sample.

**APA template:** The model fit well (CFI=1.00, RMSEA=.01, SRMR=.03); the path from *esg* to *intent* gave  $\beta=.43$ ,  $p < .001$ .

**Statistical basis:** Model estimation (lavaan) (Rosseel, Y. (2012). "lavaan: An R Package for Structural Equation Modeling." Journal of Statistical Software, 48(2), 1-36.); Fit-index cutoffs (CFI/TLI  $\geq .95$ , RMSEA  $\leq .06$ , SRMR  $\leq .08$ ) (Hu, L., Bentler, P. M. (1999). "Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives." Structural Equation Modeling, 6(1), 1-55.); Cutoffs are guidelines, not pass/fail gates (Marsh, H. W., Hau, K. T., Wen, Z. (2004). "In search of golden rules: Comment on hypothesis-testing approaches to setting cutoff values for fit indices." Structural Equation Modeling, 11(3), 320-341.); RMSEA 90% confidence interval (MacCallum, R. C., Browne, M. W., Sugawara, H. M. (1996). "Power analysis and determination of sample size for covariance structure modeling." Psychological Methods, 1(2), 130-149.); AVE / Fornell-Larcker discriminant validity (Fornell, C., Larcker, D. F. (1981). "Evaluating structural equation models with unobservable variables and measurement error." Journal of Marketing Research, 18(1), 39-50.); Reliability preference ( $\omega$  over  $\alpha$ ) (McNeish, D. (2018). "Thanks coefficient alpha, we'll take it from here." Psychological Methods, 23(3), 412-433.); Reporting standard (APA JARS-Quant) (Appelbaum, M., Cooper, H., Kline, R. B., Mayo-Wilson, E., Nezu, A. M., Rao, S. M. (2018). "Journal article reporting standards for quantitative research in psychology." American Psychologist, 73(1), 3-25.); Double-mean-centering for latent moderation (semTools::indProd) (Lin, G. C., Wen, Z., Marsh, H. W., Lin, H. S. (2010). "Structural equation models of latent interactions: Clarification of orthogonalizing and double-mean-centering strategies." Structural Equation Modeling, 17(3), 374-391.); Mean-centering strategy for latent interactions (precursor) (Marsh, H. W., Wen, Z., Hau, K. T. (2004). "Structural equation models of latent interactions: evaluation of alternative estimation strategies and indicator construction." Psychological Methods, 9(3), 275-300.); Simple slopes at -1SD/mean/+1SD (Aiken, L. S., West, S. G. (1991). Multiple Regression: Testing and Interpreting Interactions. Sage.). Full references in the exported CITATIONS.txt.

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